Python Activity-Based Learning
42. Student Marks Matrix Analysis

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

A class has marks for three students in three subjects. Store the marks in a nested list. Find each student's total and the class average for each subject.

2. Objectives

• Store marks in a nested list.

• Calculate each student's total marks.

• Calculate the average mark in each subject.

3. Assumptions

• There are three students and three subjects.

• Each mark is between 0 and 100.

• Rows represent students and columns represent subjects.

4. Algorithm (Step by Step)

1. Start.

2. Create a nested list containing marks of three students.

3. For each student, add the three subject marks and display the total.

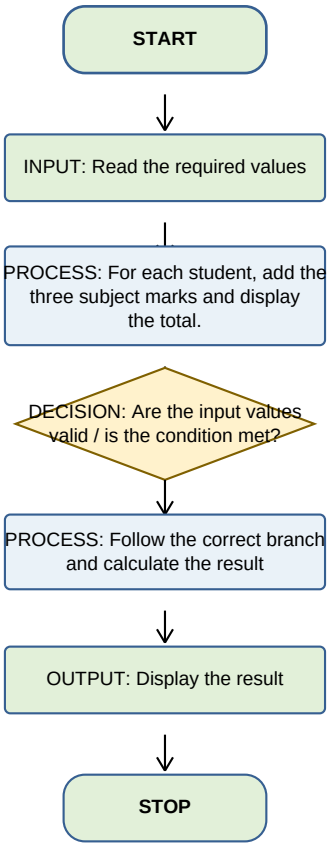
4. For each subject, add the marks of all students and divide by three.

5. Display the subject averages.

6. Stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
marks = [[80, 70, 90], [60, 75, 85], [90, 80, 95]]
FOR each student row
    total = sum of row
    PRINT total
FOR each subject column
    average = sum of that column / 3
    PRINT average
STOP
```

7. Python Source Code

```
marks = [
    [80, 70, 90],
    [60, 75, 85],
    [90, 80, 95]
]

for i in range(len(marks)):
    total = sum(marks[i])
    print("Student", i + 1, "total:", total)

for j in range(3):
    subject_total = 0
    for i in range(3):
        subject_total += marks[i][j]
    average = subject_total / 3
    print("Subject", j + 1, "average:", round(average, 2))
```

8. Test Cases

Input / Test	Expected result	What it checks
Student 1: 80,70,90	Total 240	Row total
Subject 1: 80,60,90	Average 76.67	Column average
All listed marks	Totals 240,220,265; averages 76.67,75,90	Complete matrix

9. Sample Output

```
Student 1 total: 240
Student 2 total: 220
Student 3 total: 265
Subject 1 average: 76.67
Subject 2 average: 75.0
Subject 3 average: 90.0
```

10. Conclusion

This program uses a nested list and nested loops to calculate student totals and subject averages.